Prostate Cancer Theranostics
PSMA Targeted Therapy

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KEYWORDS

• Prostate cancer • Theranostics • Prostate-specific membrane antigen • PSMA

KEY POINTS

- Prostate cancer theranostics has significantly changed the management of patients with prostate cancer.
- Prostate-specific membrane antigen–targeted radioligand therapy has shown promising results in patients with advanced metastasized castration-resistant prostate cancer.
- Clinical phase II and III trials are currently running globally to bring this new therapeutic to approval.

INTRODUCTION

Prostate cancer is one of the leading causes of cancer-related death in men.[1] After radical prostatectomy or external beam radiation therapy as the main primary therapies, the blood tumor marker prostate-specific antigen (PSA) level shall drop below detectability or to low levels, respectively. Both local and systemic therapeutic options are present in case of increasing PSA levels (which is termed biochemical recurrence). Systemic therapy options include androgen deprivation therapy (ADT), which was first proposed by Huggins who received the Nobel Prize in medicine in 1966 for this discovery. However, conventional ADT generally does not remain effective forever, and after a while, the prostate cancer ultimately becomes resistant to this type of treatment. This status is called castration resistance. Metastasized castration-resistant prostate cancer (mCRPC) is a challenging social and economic burden with the need for effective therapeutic options and individualized therapy sequences.

Since 2004, various agents have been approved for the treatment of mCRPC that prolonged overall survival and progression-free survival (PFS).[2–9] The treatment of patients with those agents depends on the metastatic tumor burden, prior performed therapies and the expected toxicity profile.[10]

Abiraterone or enzalutamide represent second-generation ADT substances that are used to treat patients who became resistant to first-generation ADT. In case of treatment failure of abiraterone or enzalutamide, the change to the other therapeutic agent is associated with a significantly lower response rate compared with the initial usage of the agent (only 8% and 18% response rate after conversion).[11,12] After resistance to abiraterone or enzalutamide, chemotherapy with docetaxel is recommended, if the performance status of the patient is sufficient.[13] Cabazitaxel is used as second-line chemotherapy. In case of symptomatic bone metastases without visceral metastases, radium-223 has been approved since 2013.[6] However, higher mortality and bone fracture rates have been reported in therapy regimes combining radium-223 with abiraterone.[14]

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Overview of Prostate-Specific Membrane Antigen–Targeted Therapy

Recently, prostate-specific membrane antigen (PSMA) targeting small molecule ligands have shown promising results in imaging and as a therapeutic option in patients with prostate cancer.[15–24] Both the lutetium-177–labeled PSMA-617 as well as PSMA-I&T (Lu-PSMA) has shown promising results in patients with mCRPC. The largest patient cohort reported in a German multicenter study using PSMA-617 comprised 145 patients and showed a biochemical response rate in 45% of patients (PSA decline of 50% and more according to prostate cancer work group III criteria).[25] Grade 3 and 4 anemia, thrombocytopenia, and leukopenia were reported in 10%, 4%, and 3%, respectively. A dry mouth was reported in 8% of the patients. Only grade 1/2 nephrotoxicity was reported (in 12% of patients). Grade 3/4 liver toxicity or nausea were not observed.

Biochemical response was associated with longer overall survival in a group of 104 patients with end-stage mCRPC receiving 177Lu-PSMA-617 radioligand therapy. This study revealed that a PSA decline of 20.87% and more was associated with a significantly longer overall survival of 16 versus 11 months ($P < .001$), which remained an independent prognosticator in a multivariate analysis.[24]

An Australian phase II study reported a biochemical response in 57% of the patients, which was higher than those previously observed in retrospective studies.[26] This increase might be because the patients were selected based on PSMA positivity assessed not only by PSMA PET but also by F-18 fluorodeoxyglucose (18F-FDG) PET computed tomography (CT). Patients with mismatch findings (positive in 18F-FDG PET but negative in PSMA PET), which might be caused by tumor dedifferentiation, were not included in the study.

The ligand PSMA I&T can be used both for imaging and therapy for patients with prostate cancer. When used as 177Lu-PSMA-I&T, it shows biochemical response in 38% of patients.[27] Grade 3/4 anemia was observed in 10%, thrombocytopenia in 4%, and neutropenia in 6%.

Besides from the beta radiation emitting lutetium-177, alpha radiation emitting actinium-225 has also been used for PSMA-targeted therapy. Actinium-225–labeled PSMA-617 showed a biochemical response in 63% of patients with a tumor control duration of 9 months in median.[28] Tumor control was denoted the time from the start of PSMA therapy till changed treatment regime. Actinium-225 as an alpha emitter might be more effective than lutetium-177 but has a poorer toxicity profile and comparative data confirming its purported efficacy is lacking. In particular, permanent xerostomia was a main cause for therapy termination.[29] Xerostomia might have a higher frequency in actinium-225 ligands due to alpha radiation. Therefore, future developments of PSMA ligands should therefore focus on reducing the affinity to salivary glands to facilitate a low toxicity profile.

Overview of Recent and Ongoing Prospective Prostate-Specific Membrane Antigen Trials

An overview of the current running studies using radiolabeled PSMA-617 is given in **Table 1**.

The entire prostate cancer therapy community is eagerly awaiting the results of the phase III VISION trial.[30] In this open-label prospective study patients are randomized on a 2:1 basis to receive Lu-PSMA plus standard of care (SOC) therapy or SOC alone with overall survival and radiological PFS as endpoints. Patients must have received at least one line of chemotherapy, but no more than 2, and must have had received at least one of the second-generation ADT abiraterone or enzalutamide. The results of this study will hopefully lead to the approval of Lu-PSMA and thereby make it available for larger group of patients.

Initial results of TheraP, a randomized phase II trial of Lu-PSMA radioligand therapy versus cabazitaxel in mCRPC progressing after docetaxel, were presented at annual meeting of the society of clinical oncology in 2020.[31] In the presented analysis, 98 patients had received Lu-PSMA and 85 patients were treated with cabazitaxel. Lu-PSMA-treated patients showed a higher rate of biochemical response than those treated with cabazitaxel (67% vs 33%). In addition, total rates of grade 3/4 hematoxicity were lower in patients treated with Lu-PSMA compared with cabazitaxel (35 % vs 54%). A phase III trial implementing this protocol is needed to confirm these results.[31]

IMAGING AND THERAPY

With the rise of advanced image analysis techniques, semiautomated approaches for the delineation of all metastases in PSMA PET/CT have been proposed.[32,33] It was shown that the comprehensive analysis of PSMA expression in all metastases could identify patients with favorable, intermediate, and poor outcome.[34] Interestingly, patients with best outcome had exclusively metastases with strong PSMA expression, whereas patients with side by side presence of strong and weak PSMA expression had intermediate outcome. Patients with a generally low PSMA expression had poor outcome. This might be

Table 1
Overview of the current running prospective studies using ^{177}Lu-PSMA-617

Study (NCT) n (Number of Patients)	Phase	Study Description	End Points
LuPARP NCT03874884 n = 52	I	^{177}Lu-PSMA-617 therapy and olaparib in patients with metastatic castration-resistant prostate cancer	Dose-limiting toxicities
NCT03042468 n = 48	I	Phase I dose-escalation study of fractionated 177Lu-PSMA-617 for progressive metastatic CRPC	Dose-limiting toxicities
PRINCE NCT03658447 n = 37	Ib/II	Phase Ib/II study of radionuclide ^{177}lutetium-PSMA-617 therapy in combination with pembrolizumab for treatment of metastatic castration-resistant prostate cancer (mCRPC)	Safety PSA response
NCT0354165 n = 48	Ib/II	^{177}Lu-J591 and 177Lu-PSMA-617 combination for mCRPC	Dose-limiting toxicities PSA response
TheraP NCT03392428 n = 200	II	A randomized phase II trial of ^{177}Lu-PSMA-617 (Lu-PSMA) theranostic vs cabazitaxel in metastatic castration resistant prostate cancer (mCRPC) progressing after docetaxel	PSA response rate Safety
VISION NCT03511664 n = 750	III	An international, prospective, open-label, multicenter, randomized phase III study of ^{177}Lu-PSMA-617 in the treatment of patients with progressive PSMA-positive metastatic castration-resistant prostate cancer	Overall survival

explained by tumor dedifferentiation: in the course of the disease, the expression of the PSMA molecule is lost, whereas neuroendocrine cell markers are gained.[35,36] Thereby, the appearance of metastases with low PSMA expression might resemble the presence of dedifferentiated prostate cancer cells. However, an alternative explanation might be that Lu-PSMA might deliver more dosage to metastases with strong PSMA expression, which in turn results in favorable outcome. Future studies are needed to elucidate these initial findings.

There are divergent reports on the prognostic value of the baseline total tumor volume. Ferdinandus and colleagues[37] have stated that the PSMA PET–derived tumor volume is not a significant predictor of overall survival time. In contrast, Seifert and colleagues[38,39] could show that the total tumor volume is a significant predictor of overall survival in a single center analysis, which was corroborated in a bicentric analysis. Currently used dose regimes are not tumor volume specific. The concentration of the administered therapeutic ligand is therefore dependent on the total tumor volume, which is sometimes denoted as tumor sink-in effect. Therefore, it seems plausible that the total tumor volume is a negative prognosticator of overall survival, as the maximal dose delivered through Lu-PSMA should depend on the ratio of applied dose per tumor volume. This thought is underlined by the fact that the metric total lesion quotient (denoted in analogy to the FDG PET metric total lesion glycolysis), which is obtained by dividing the tumor volume by the medium PSMA uptake, is an even better prognosticator of overall survival time.[39] Therefore, future studies should evaluate this novel parameter further.

FUTURE DIRECTIONS

Future directions of PSMA-targeted therapy will likely focus on combinational therapies, dose escalation, personalized doses, and on optimized patient selection.

A combination of PSMA-targeted radioligand therapy and second-generation ADT combined with inhibitors of double-strand DNA break repair mechanisms (poly(ADP)-ribose polymerase inhibitors) may improve therapeutic efficacy at low toxicity and should be investigated in future studies.[40]

Only a small number of studies have evaluated the relevance of the applied radiation dosage in Lu-PSMA therapy regimes. Seifert and colleagues[41] have retrospectively evaluated the response of patients treated with 7.5 GBq every 6 weeks compared with those treated with 6 GBq every 8 weeks. The median number of cycles was 4 in the 7.5 GBq group versus 3 in the 6.0 GBq group. As expected, the 7.5 GBq group showed better response to therapy, which was however not statistically significant. Rathke and colleagues[42] have applied single doses of up to 9.3 GBq. However, to date there is no systematic analysis whether higher doses result in better outcome.

All retrospective and prospective studies have one thing in common: it is still not known why some patients respond and some patients do not respond to PSMA-targeted therapy. This might be partly explained by the heterogeneity of prostate cancer, especially in later stages: it is a rather complex disease with various classification and gradings systems already at the beginning of the disease, for example, Gleason score, initial PSA, and others. Moreover, prostate cancer lesions are heterogeneous within the same patient, which becomes visible by PSMA-targeted imaging showing different PSMA expression in different metastatic lesions. This heterogeneity of PSMA expression might be a predictor for efficacy of PSMA-targeted therapy. The tumor and its metastases undergo an evolutionary process within the course of disease, which is also influenced by the therapies performed. Although the topic of different tumor types and clones is complex, currently only the presence of liver metastases is known to be a negative predictor for outcome of patients treated with Lu-PSMA therapy.[43] Therefore, precision medicine might be needed to predict the response of a patient with prostate cancer to Lu-PSMA by analyzing patient-specific tumor characteristics. To give an example, several studies could show that androgen receptor splice variant 7 in circulating tumor cells is a negative predictor of response to abiraterone and enzalutamide because of the lack of the ligand-binding domain on tumor cells.[44] To unleash the full potential of radionuclide therapy in prostate cancer, those kinds of studies are needed for patients receiving Lu-PSMA as well.

DISCLOSURES

KR has received consultant fees from Bayer and ABX; lectureship fees from Janssen Cilag, Amgen, AAA, and SIRTEX; and has received travel expenses from Endocyte as an unpaid member of the steering committee of the VISION phase III trial.

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